

```
import pandas as pd
import numpy as np
df=pd.read_csv("house_prices_practice.csv")
```

```
display(df)
```

	Id	OverallQual	GrLivArea	GarageCars	TotalBsmtSF	YearBuilt
0	1	7	1560	0	1658	1969
1	2	4	2827	2	1319	2012
2	3	8	3920	0	841	2010
3	4	5	3044	0	1058	1998
4	5	7	801	1	2428	2020
..	...	...	...	...	...	...
295	296	1	3495	1	1792	1954
296	297	5	3438	3	1266	2003
297	298	6	1992	0	1148	1996
298	299	3	3722	1	1407	1998
299	300	9	1667	0	988	1958

	FullBath	BedroomAbvGr	LotArea	SalePrice
0	2	1	8059	177106
1	3	4	13530	301044
2	1	4	9010	360609
3	0	4	13207	240556
4	0	1	9117	193656
..	...	...	...	...
295	2	5	4978	250604
296	0	1	9373	329906
297	1	1	7907	184623
298	1	1	8097	303345
299	2	1	14553	204982

```
[300 rows x 10 columns]
```

```
df.head()
```

	Id	OverallQual	GrLivArea	GarageCars	TotalBsmtSF	YearBuilt
0	1	7	1560	0	1658	1969

```

2
1 2 4 2827 2 1319 2012
3
2 3 8 3920 0 841 2010
1
3 4 5 3044 0 1058 1998
0
4 5 7 801 1 2428 2020
0

```

```

    BedroomAbvGr  LotArea  SalePrice
0              1    8059    177106
1              4   13530    301044
2              4    9010    360609
3              4   13207    240556
4              1    9117    193656

```

```
df.shape
```

```
(300, 10)
```

```
df.info()
```

```

<class 'pandas.DataFrame'>
RangeIndex: 300 entries, 0 to 299
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Id                    300 non-null   int64
1   OverallQual           300 non-null   int64
2   GrLivArea             300 non-null   int64
3   GarageCars            300 non-null   int64
4   TotalBsmtSF           300 non-null   int64
5   YearBuilt             300 non-null   int64
6   FullBath              300 non-null   int64
7   BedroomAbvGr          300 non-null   int64
8   LotArea               300 non-null   int64
9   SalePrice             300 non-null   int64

```

```
dtypes: int64(10)
```

```
memory usage: 23.6 KB
```

```
df.isnull().sum()
```

```

Id              0
OverallQual     0
GrLivArea       0
GarageCars      0
TotalBsmtSF     0
YearBuilt       0
FullBath        0
BedroomAbvGr    0

```

```
LotArea      0
SalePrice    0
dtype: int64
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df.describe()
```

	Id	OverallQual	GrLivArea	GarageCars	
TotalBsmtSF \					
count	300.000000	300.000000	300.000000	300.000000	300.000000
mean	150.500000	5.326667	2307.386667	1.330000	1468.796667
std	86.746758	2.873001	1042.561303	1.109898	672.333705
min	1.000000	1.000000	504.000000	0.000000	303.000000
25%	75.750000	3.000000	1392.250000	0.000000	903.000000
50%	150.500000	5.000000	2265.500000	1.000000	1502.000000
75%	225.250000	8.000000	3306.500000	2.000000	2129.500000
max	300.000000	10.000000	3998.000000	3.000000	2492.000000

	YearBuilt	FullBath	BedroomAbvGr	LotArea	
SalePrice					
count	300.000000	300.000000	300.000000	300.000000	300.000000
mean	1986.163333	1.523333	2.926667	8969.453333	252262.903333
std	21.377089	1.131543	1.456604	3753.531132	74998.055214
min	1950.000000	0.000000	1.000000	2009.000000	82494.000000
25%	1967.000000	0.000000	2.000000	5996.250000	190355.250000
50%	1986.000000	2.000000	3.000000	9031.000000	251292.500000
75%	2004.250000	3.000000	4.000000	12316.000000	307105.000000
max	2023.000000	3.000000	5.000000	14987.000000	435291.000000

```
df.drop_duplicates()
```

	Id	OverallQual	GrLivArea	GarageCars	TotalBsmtSF	
YearBuilt \						
0	1	7	1560	0	1658	1969
1	2	4	2827	2	1319	2012
2	3	8	3920	0	841	2010
3	4	5	3044	0	1058	1998
4	5	7	801	1	2428	2020
..	...	...	...	...	...	...
295	296	1	3495	1	1792	1954
296	297	5	3438	3	1266	2003
297	298	6	1992	0	1148	1996
298	299	3	3722	1	1407	1998
299	300	9	1667	0	988	1958

	FullBath	BedroomAbvGr	LotArea	SalePrice
0	2	1	8059	177106
1	3	4	13530	301044
2	1	4	9010	360609
3	0	4	13207	240556
4	0	1	9117	193656
..	...	...	...	...
295	2	5	4978	250604
296	0	1	9373	329906
297	1	1	7907	184623
298	1	1	8097	303345
299	2	1	14553	204982

[300 rows x 10 columns]

```

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_squared_error

X = df.drop(['Id', 'SalePrice'], axis=1)
y = df['SalePrice']

X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42)

```

```

model = LinearRegression()
model.fit(X_train, y_train)

LinearRegression()

print("\nIntercept:", model.intercept_)
for col, coef in zip(X.columns, model.coef_):
    print(f"Coefficient for {col}: {coef:.4f}")

Intercept: -1512820.5684315271
Coefficient for OverallQual: 11914.6881
Coefficient for GrLivArea: 55.4103
Coefficient for GarageCars: 19109.6416
Coefficient for TotalBsmtSF: 25.1990
Coefficient for YearBuilt: 746.3710
Coefficient for FullBath: 11040.3430
Coefficient for BedroomAbvGr: 4812.0620
Coefficient for LotArea: -0.1595

y_pred = model.predict(X_test)
r2 = r2_score(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))

print(f"\nR2 Score: {r2:.4f}")
print(f"Root Mean Squared Error (RMSE): ${rmse:,.2f}")

R2 Score: 0.9724
Root Mean Squared Error (RMSE): $12,505.46

```